Davide Sclosa

Postdoctoral Researcher in Mathematics, King's College London

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Research Interests

Ergodic theory, dynamical systems, coupled maps, ergodic Ramsey theory, combinatorics, graph limits.

Positions

Postdoctoral Researcher Nov 2025 – present
King's College London, UK
Coupled maps on graph limits and action convergence of nonlinear graph operators. With Matteo Tanzi.

Postdoctoral Researcher Nov 2024 – Nov 2025
University of Crete, Greece
Ergodic theory with applications to additive combinatorics and number theory. With Nikos Frantzikinakis.

Education

PhD in Mathematics Nov 2020 – Feb 2025
Vrije Universiteit Amsterdam, Netherlands
Thesis: *Dynamical Systems on Finite Graphs and Graph Limits*. Advisor: Christian Bick.

MSc in Mathematics Dec 2018 – Jul 2020
University of Udine, Italy
110/110 cum laude. Thesis on hyperbolic billiards and combinatorics on words, advisor Giovanni Panti; led to [11].

BSc in Mathematics Sep 2015 – Dec 2018
University of Udine, Italy
110/110 cum laude. Thesis on algebraic groups over finite fields, advisor Pietro Corvaja; led to [10].

Publications

Single-authored unless co-authors are listed.

Preprints

[1] **Multidimensional Szemerédi theorem for pairwise independent jointly intersective polynomials**
with Nikos Frantzikinakis. arXiv:2610.18657, 2026.

[2] **Dimension bounds for systems of equations with graph structure**
with Eddie Nijholt. arXiv:2410.06840, 2024. Submitted.

[3] **A random-walk concentration principle for occupancy processes on finite graphs**
with Michel Mandjes and Christian Bick. arXiv:2410.06807, 2024. Submitted.

Journal articles

[4] **If you are the smartest person in the room, you are in the wrong room**
Graphs and Combinatorics 41(5), 2025.

[5] **From combinatorics to geometry: the dynamics of graph gradient diffusion**
Geometriae Dedicata 219(1), 2025.

[6] **Bounded power series on the real line**
Journal of Mathematical Analysis and Applications 538(2), 2024.

[7] **Dynamical systems on graph limits and their symmetries**
with Christian Bick. *Journal of Dynamics and Differential Equations*, 2024.

[8] **Completely degenerate equilibria of the Kuramoto model on networks**
Nonlinearity 37(9), 2024.

[9] **Kuramoto networks with infinitely many stable equilibria**
SIAM Journal on Applied Dynamical Systems 22(4), 2023.

[10] **Algebraic groups over finite fields: connections between subgroups and isogenies**
Journal of Group Theory 26(6), 1143–1155, 2023.

[11] **The finiteness conjecture holds in $\mathrm{SL}_2(\mathbb{Z}_{\geq 0})$**
with Giovanni Panti. *Nonlinearity* 34(8), 5234, 2021.

Invited Talks

- ETDS Seminar, *University of Warwick, UK* Nov 2026
- 13th Miniworkshop on Operator Theoretic Aspects of Ergodic Theory, *Kiel, Germany* Oct 2026
- Analysis Seminar, *University of Crete, Greece* Dec 2024
- NetDyn2024 Workshop, *Lund University, Sweden* Aug 2024
- One World Dynamics Seminar, *online* Jun 2024
- Networks Seminar, *University of Oxford, UK* Apr 2024
- Dynamical Systems Seminar, *Imperial College London, UK* Mar 2024
- Dynamics Seminar, *University of Exeter, UK* Mar 2024
- Floris Takens Seminar, *University of Groningen, Netherlands* Sep 2023
- Dynamics Days Europe, *Naples, Italy* Sep 2023
- SIAM Conference on Applications of Dynamical Systems, *Portland, USA* May 2023
- Oberseminar Dynamics, *TU Munich, Germany* Sep 2021
- Lowlands Dynamics Seminar, *online* Apr 2021

Selected Conferences and Schools Attended

- Perspectives on Ergodic Theory and its Interactions, celebrating Vitaly Bergelson, *IMPAN, Warsaw, Poland* Jul 2025
- Summer School on Additive Combinatorics, Number Theory, and Ergodic Theory, *EPFL, Switzerland* Jul 2025
- Dynamics Days Europe, *Bremen, Germany* Jul 2024
- Hypergraphs: Theory and Applications, *Isaac Newton Institute, UK* Jul 2024

Teaching, Mentoring, and Service

- **Teaching assistant**, Vrije Universiteit Amsterdam, 2020–2024. Dynamical Systems, Discrete Mathematics, Mathematical Analysis, Mathematical Modeling.
- **Mentoring**: co-supervision of a Bachelor's thesis, Vrije Universiteit Amsterdam.
- **Referee** for *Transactions of the American Mathematical Society*; *Forum of Mathematics, Sigma*; *Journal of Nonlinear Science*; *SIGMA (Symmetry, Integrability and Geometry: Methods and Applications)*; *Chaos*.
- **Workshops organized**: *Graph Stochastic Dynamics and Dynamic Stochastic Graphs* (2024); *Dynamics of Networks with Switching Topology* (2023).
- **Hiring committees**: tenure-track position (2022), PhD position (2024).
- **Industry**: Study Group Mathematics with Industry, Groningen, 2023; report in *Mathematics in Industry Reports*, 2024.